

CURRICULUM VITAE

Family name, First name: Magyar, Norbert

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• EDUCATION

2014/10 – 2018/10: PhD in Mathematics, Title: “*Back-reaction of the plasma on coronal oscillations*”

Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium

Name of PhD Supervisor: Prof. Tom Van Doorselaere

2012/10 – 2014/07: Master in Computational Physics

Faculty of Physics, “Babeş-Bolyai” University, Cluj-Napoca, Romania

• CURRENT POSITION

2026/09 – Space Data Solutions Architect
Antwerp Space, Belgium

• PREVIOUS POSITIONS

2024/12 – 2026/09 Space Data Software Engineer
Starion Technologies N.V., Diegem, Belgium

2021/07 – 2024/07 FWO Postdoctoral Fellow
Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium

2019/07 – 2021/07: Newton International Fellow of the Royal Society
Centre for Fusion, Space, and Astrophysics (CFSA), Dept. of Physics, University of Warwick, UK

2018/10 – 2019/07 Postdoctoral mandate of KU Leuven
Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium

• FELLOWSHIPS AND AWARDS

I have been self-financed through awarded fellowships my whole research career.

- 2020 FWO Postdoctoral Fellowship (3 years, 2020 success rate: ~25%)
- 2019 Newton International Fellowship of the Royal Society
(2 years, 2019 success rate: ~8%, Award: ~84k€)
- 2019 **PhD Thesis Prize**, The European Solar Physics Division (ESPD) of the European Physical Society (EPS)
“Prize awarded for significant contributions, using 3D magneto-hydrodynamics numerical experiments, to the study of waves and their relation to turbulence in the solar corona”
- 2018 Postdoctoral mandate (PDM) (1 year, success rate ~30%)
- 2014 FWO PhD Fellowship (4 years, 2014 success rate ~22%)

• TEACHING ACTIVITIES

- 2021 – 2022 Teaching assistant – Plasma Physics, Dept. of Mathematics, KU Leuven, Belgium
- 2020 – 2021 **Guest lecturer** – Solar Physics, University of Warwick, UK
- 2016 – 2019 Teaching assistant – Fluid Dynamics, Dept. of Mathematics, KU Leuven, Belgium
- 2015 – 2018 Teaching assistant – Space Weather, Dept. of Mathematics, KU Leuven, Belgium

• ORGANISATION OF SCIENTIFIC MEETINGS

- 2022 **ISSI (International Space Science Institute) Team Leader**, Title: *“Turbulence at the Edge of the Solar Corona: Constraining Available Theories Using the Latest Parker Solar Probe Measurements”*, 15 team members in two one-week meetings, Bern, Switzerland
- 2022 Member of SOC, LOC, CmPA Conference on Solar Physics, ~70 participants, KU Leuven, Belgium
- 2021 Convener of session on solar coronal heating mechanisms, National Astronomy Meeting (NAM) of the UK, ~60 participants in the session, University of Bath, UK

• INSTITUTIONAL RESPONSIBILITIES

- 2022 – 2023 Master student Advisor (Willem Keppens), Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium
- 2021 – 2024 **PhD Student Co-Advisor** (Rajab Ismayilli), Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium
- 2020 – 2021 PhD Student Supervisory Committee Member (Rajab Ismayilli), Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium
- 2019 Master Internship Advisor (Pauline Simon), Centre for mathematical Plasma Astrophysics (CmPA), Dept. of Mathematics, KU Leuven, Belgium

• REVIEWING ACTIVITIES

- 2020 Reviewer, ASCR Leadership Computing Challenge (ALCC) proposals, DOE, US

2017 – 2024 Reviewer for The Astrophysics Journal, Nature, MNRAS, A&A, Solar Physics

- **MEMBERSHIPS OF SCIENTIFIC SOCIETIES**

2019 – IAU Junior Member

- **MAJOR COLLABORATIONS**

2022 **Two-month research stay at NASA/GSFC, USA.**
Collaborators: Adam Szabo (Parker Solar Probe Mission Scientist), Jaye Verniero;
Topic: Parker Solar Probe data analysis.

2020 – 2021 Collaboration with Robert Erdélyi, University of Sheffield, UK
Topic: Magnetic switchback generation

2019 – 2020 ISSI Beijing Workshop. Collaborative effort resulting in multiple co-authored, invited Space Science Reviews publications on solar coronal physics.

2017 GREGOR Solar observation campaign. Collaborator: Dominik Utz
Topic: Solar Photospheric observations, Teide Observatory, Spain

2015 Collaboration with Jaume Terradas, University of Balearic Islands, Spain
Topic: Simulations of coronal loop oscillations.

- **INVITED TALKS AND SEMINARS**

2015 Invited Early-Career Researcher on ISSI Team ‘Large-Amplitude Oscillations as a Probe of Quiescent and Erupting Solar Prominences’, Bern, Switzerland.

2017-2018 Invited Early-Career Researcher on ISSI Team ‘Towards Dynamic Solar Atmospheric Magneto-Seismology with New Generation Instrumentation’, Bern, Switzerland.

2018 Invited talk at BUKS Workshop ‘Waves and Instabilities in the Solar Atmosphere: Confronting the Current State-of-the Art’, La Laguna, Spain.

2019 Seminar of the Solar Physics group, University of Northumbria, UK.

2019 Seminar at the Plasma Dynamics Group, University of Sheffield, UK.

2019 Seminar at Center for Fusion, Space, and Astrophysics, University of Warwick, UK.

2019 Invited team member in ISSI Beijing Workshop, ‘Oscillatory Processes in Solar and Stellar Coronae’, Beijing, China.

2020 PhD Prize Talk at the European Solar Physics Division conference (cancelled due to Covid).

2021 Seminar at Solar and Magnetospheric Theory Group, University of St. Andrews, UK (virtual).

2021 Invited talk at AOGS conference, session on Solar Physics (virtual).

2022 Seminar at NASA/GSFC Heliospheric Physics Laboratory, Greenbelt, USA.

2022 Invited talk at the 4th Workshop on Magnetic Fields in Laboratory High Energy Density Plasmas (LaB), Paris, France.

2023 Invited review on “Alfvén waves in the solar atmosphere”, AGU Chapman conference, Berlin, Germany.

2023 Invited workshop member at ISSI: ‘Magnetic Switchbacks in the Young

2024	Solar Wind', Bern, Switzerland Invited working group member at ISSI: 'Why Do We Still Have a "Coronal Heating Problem" 8 Decades After Edlen's 1943 Article? ', Bern, Switzerland
2026	Talk at ESA's Space Weather Service Network Workshop, ESOC, Darmstadt, Germany
2026	"Architecture and Innovations for ESA's Space Weather Payload Data Centre" at ESAW 2026, ESA-ESOC, Darmstadt, Germany

PUBLICATION LIST

According to NASA/ADS, as of October 2026, I have co-authored 39 peer-reviewed publications, 15 of which as the PI or first author. Some other important metrics are the following:

- 1256 citations (395 for first-authored papers)
- H-Index of 19 (12 for first-authored papers)
- 10 papers without the PhD supervisor, T. Van Doorselaere (5 first-authored papers)

First-authored papers:

- 1) Magyar, N., Verniero, J., Szabo, A., Zhang, J., Van Doorselaere, T.; "*Solar wind data analysis aided by synthetic modeling: A better understanding of plasma frame variations from temporal data*", Astronomy & Astrophysics, Volume 688, id.A74, 10 pp., 2024.
- 2) Magyar, N.; "*Alfvén Waves in the Solar Atmosphere*", Alfvén Waves Across Heliophysics: Progress, Challenges, and Opportunities, Edited by Andreas Keiling. ISBN: 9781394195954. Geophysical Monograph Series, Vol. 285. Wiley, p.3-38, 2024.
- 3) Magyar, N., Van Doorselaere, T.; "*Phase Mixing and the $1/f$ Spectrum in the Solar Wind*", The Astrophysical Journal, Volume 938, Issue 2, id.98, 9 pp., 2022.
- 4) Magyar, N., Duckenfield, T., Van Doorselaere, T., Nakariakov, V. M.; "*The dynamics and observability of circularly polarized kink waves*", Astronomy & Astrophysics, Volume 659, id.A73, 8 pp., 2022.
- 5) Magyar, N., Utz, D., Erdélyi, R., Nakariakov, V. M.; "*Could Switchbacks Originate in the Lower Solar Atmosphere? II. Propagation of Switchbacks in the Solar Corona*", The Astrophysical Journal, Volume 914, Issue 1, id.8, 13 pp., 2021.
- 6) Magyar, N., Utz, D., Erdélyi, R., Nakariakov, V. M.; "*Could Switchbacks Originate in the Lower Solar Atmosphere? I. Formation Mechanisms of Switchbacks*", The Astrophysical Journal, Volume 911, Issue 2, id.75, 7 pp., 2021.

- 7) Magyar, N., Nakariakov, V. M.; “*Three-dimensional Simulations of the Inhomogeneous Low Solar Wind*”, The Astrophysical Journal, Volume 907, Issue 1, id.55, 16 pp., 2021.
- 8) Magyar, N., Nakariakov, V. M.; “*Standing Kink Waves in Sigmoid Solar Coronal Loops: Implications for Coronal Seismology*”, The Astrophysical Journal Letters, Volume 894, Issue 2, id.L23, 5 pp., 2020.
- 9) Magyar, N., Van Doorselaere, T., Goossens, M.; “*Understanding Uniturbulence: Self-cascade of MHD Waves in the Presence of Inhomogeneities*”, The Astrophysical Journal, Volume 882, Issue 1, article id. 50, 11 pp., 2019.
- 10) Magyar, N., Van Doorselaere, T., Goossens, M.; “*The Nature of Elsässer Variables in Compressible MHD*”, The Astrophysical Journal, Volume 873, Issue 1, article id. 56, 7 pp., 2019.
- 11) Magyar, N., Van Doorselaere, T.; “*Assessing the Capabilities of Dynamic Coronal Seismology of Alfvénic Waves through Forward Modeling*”, The Astrophysical Journal, Volume 856, Issue 2, article id. 144, 9 pp., 2018.
- 12) Magyar, N., Van Doorselaere, T., Goossens, M.; “*Generalized phase mixing: Turbulence-like behaviour from unidirectionally propagating MHD waves*”, Scientific Reports, Volume 7, id. 14820, 2017.
- 13) Magyar, N., Van Doorselaere, T.; “*Damping of nonlinear standing kink oscillations: a numerical study*”, Astronomy & Astrophysics, Volume 595, id.A81, 9 pp., 2016.
- 14) Magyar, N., Van Doorselaere, T.; “*The Instability and Non-existence of Multi-stranded Loops When Driven by Transverse Waves*”, The Astrophysical Journal, Volume 823, Issue 2, article id. 82, 8 pp., 2016.
- 15) Magyar, N., Van Doorselaere, T., Marcu, A.; “*Numerical simulations of transverse oscillations in radiatively cooling coronal loops*”, Astronomy & Astrophysics, Volume 582, id.A117, 8 pp., 2015.

Contributing author papers (excluding conf. Proc.):

- 1) Wyper, P. F., Squire, J., Pariat, E., Agapitov, O. V., Drake, J. F., **Magyar, N.**, Matthaeus, W. H., Matteini, L., Ruffolo, D., Réville, V., Shi, C., Shoda, M., Swisdak, M., Velli, M., Akhavan-Tafti, M., Gannouni, B., Lionello, R., Madjarska, M. S., Owens, M. J., Raouafi, N. E., Sterling, A. C., Tripathi, D.; “*Magnetic Switchback Formation: A Review of Proposed Mechanisms*”, Space Science Reviews, Volume 222, Issue 4, id.43, 2026.
- 2) McMurdo, M., Van Doorselaere, T., **Magyar, N.**, Banović, L., Lim, D.; “*The Uniturbulence and Alfvén Wave Solar Model (UAWSoM) in MPI-AMRVAC*”, Astronomy & Astrophysics, Volume 705, id.A15, 13 pp., 2026.

- 3) Van Doorselaere, T., Sieyra, M. V., **Magyar**, N., Goossens, M., Banović, L.; *“Uniturbulence and Alfvén wave solar model”*, Astronomy & Astrophysics, Volume 696, id.A166, 18 pp., 2025.
- 4) Scolini, C., Zhuang, B., Lugaz, N., Winslow, R. M., Farrugia, C. J., **Magyar**, N., Bacchini, F.; *“Can Alfvénic Fluctuations Affect the Correlation and Complexity of Magnetic Field of Magnetic Ejecta? A Case Study Based on Multi-spacecraft Measurements at 1 au”*, The Astrophysical Journal, Volume 978, Issue 2, id.146, 12 pp., 2025.
- 5) Sishtla, C. P., Pomoell, J., **Magyar**, N., Kilpua, E., Good, S.; *“Validity of using Elsässer variables to study the interaction of compressible solar wind fluctuations with a coronal mass ejection”*, Astronomy & Astrophysics, Volume 683, id.A171, 10 pp., 2024.
- 6) Keppens, W., **Magyar**, N., Van Doorselaere, T.; *“Magnetic power spectrum variability with large-scale total magnetic field fluctuations”*, Astronomy & Astrophysics, Volume 683, id.A114, 14 pp., 2024.
- 7) Van Doorselaere, T., **Magyar**, N., Sieyra, M. V., Goossens, M.; *“The magnetohydrodynamic equations in terms of waveframe variables”*, Journal of Plasma Physics, Volume 90, Issue 1, article id.905900113, 2024.
- 8) Scolini, C., Lugaz, N., Winslow, R. M., Farrugia, C. J., **Magyar**, N., Bacchini, F.; *“On the Role of Alfvénic Fluctuations as Mediators of Coherence within Interplanetary Coronal Mass Ejections: Investigation of Multi-spacecraft Measurements at 1 au”*, The Astrophysical Journal, Volume 961, Issue 1, id.135, 13 pp., 2024.
- 9) Sishtla, C. P., Jebaraj, I. C., Pomoell, J., **Magyar**, N., Pulupa, M., Kilpua, E., Bale, S. D.; *“The Effect of the Parametric Decay Instability on the Morphology of Coronal Type III Radio Bursts”*. The Astrophysical Journal Letters, Volume 959, Issue 2, id.L33, 9 pp., 2023.
- 10) Petrova, E., **Magyar**, N., Van Doorselaere, T., Berghmans, D.; *“High-frequency Decayless Waves with Significant Energy in Solar Orbiter/EUI Observations”*, The Astrophysical Journal, Volume 946, Issue 1, id.36, 11 pp., 2023.
- 11) Ismayilli, R., Van Doorselaere, T., Goossens, M., **Magyar**, N.; *“Non-linear damping of surface Alfvén waves due to Uniturbulence”*, Frontiers in Astronomy and Space Sciences, Volume 8, id.241, 2022.
- 12) Banerjee, D., Krishna Prasad, S., Pant, V., McLaughlin, J. A., Antolin, P., **Magyar**, N., Ofman, L., Tian, H., Van Doorselaere, T., De Moortel, I., Wang, T. J.; *“Magnetohydrodynamic Waves in Open Coronal Structures”*, Space Science Reviews, Volume 217, Issue 7, article id.76, 2021.
- 13) Nakariakov, V. M., Anfinogentov, S. A., Antolin, P., Jain, R., Kolotkov, D. Y., Kupriyanova, E. G., Li, D., **Magyar**, N., Nisticò, G., Pascoe, D. J., Srivastava, A. K., Terradas, J., Vasheghani Farahani, S., Verth, G., Yuan, D., Zimovets, I. V.; *“Kink*

Oscillations of Coronal Loops”, Space Science Reviews, Volume 217, Issue 6, article id.73, 2021.

- 14) Van Doorselaere, T., Goossens, M., **Magyar**, N., Ruderman, M. S., Ismayilli, R.; “*Nonlinear Damping of Standing Kink Waves Computed With Elsässer Variables*”, The Astrophysical Journal, Volume 910, Issue 1, id.58, 8 pp., 2021.
- 15) Van Doorselaere, T., Srivastava, A. K., Antolin, P., **Magyar**, N., Vasheghani Farahani, S., Tian, H., Kolotkov, D., Ofman, L., Guo, M., Arregui, I., De Moortel, I., Pascoe, D.; “*Coronal Heating by MHD Waves*”, Space Science Reviews, Volume 216, Issue 8, article id.140, 2020.
- 16) Pant, V., **Magyar**, N., Van Doorselaere, T., Morton, R. J.; “*Investigating “Dark” Energy in the Solar Corona Using Forward Modeling of MHD Waves*”, The Astrophysical Journal, Volume 881, Issue 2, article id. 95, 14 pp., 2019.
- 17) Krishna Prasad, S., Raes, J. O., Van Doorselaere, T., **Magyar**, N., Jess, D. B.; “*The Polytopic Index of Solar Coronal Plasma in Sunspot Fan Loops and Its Temperature Dependence*”, The Astrophysical Journal, Volume 868, Issue 2, article id. 149, 9 pp., 2018.
- 18) Terradas, J., **Magyar**, N., Van Doorselaere, T.; “*Effect of Magnetic Twist on Nonlinear Transverse Kink Oscillations of Line-tied Magnetic Flux Tubes*”, The Astrophysical Journal, Volume 853, Issue 1, article id. 35, 15 pp., 2018.
- 19) Mandal, S., **Magyar**, N., Yuan, D., Van Doorselaere, T., Banerjee, D.; “*Forward Modeling of Propagating Slow Waves in Coronal Loops and Their Frequency-dependent Damping*”, The Astrophysical Journal, Volume 820, Issue 1, article id. 13, 9 pp., 2016.
- 20) Van Doorselaere, T., Antolin, P., Yuan, D., Reznikova, V., **Magyar**, N.; “*Forward modelling of optically thin coronal plasma with the FoMo tool*”, Frontiers in Astronomy and Space Sciences, Volume 3, id.4, 2016.
- 21) Mocanu, G., **Magyar**, N., Pardi, A., Marcu, A.; “*Appearance of an accretion disc perturbed by fractional Brownian Motion density*”, Monthly Notices of the Royal Astronomical Society, Volume 439, Issue 4, p.3790-3797, 2014.

STATEMENT OF RESEARCH EXPERIENCE

I am an astrophysicist having wide-ranging interests in the field, with solar atmospheric and heliospheric plasma physics as my principal areas of expertise. I am skilled in numerical and analytical work, applied to linear and nonlinear MHD waves, their dissipation and heating, MHD turbulence, instabilities, forward-modeling of EUV emission. To a lesser degree, I have participated in and carried out observational work, using the GREGOR, VTT telescopes, and SDO/AIA satellite. I have also undertaken analysis of in-situ

data from the Parker Solar Probe, with the aim of comparing my previous results from 3D full MHD simulations of the solar corona and solar wind. Besides these areas of research in which I have published, I am very open and willing to work on other adjacent topics in the future, such as the subject of this position, atmospheric evolution. I am always tending to broaden my scientific horizons and to acquire a more holistic understanding of astrophysics and related areas, including the physics of the solar system.

I have started my research career studying oscillations of solar coronal loops via numerical simulations. I believe that my research independence and versatility are clearly reflected in my initiative, shortly after the start of my PhD, to lead and participate in theoretical and observational works besides numerical ones, different than my PhD supervisor's research. Being self-funded since the start of my research career through various fellowships, I have transitioned to doing research on the topics of my own initiative, such as the basics of MHD turbulence and solar wind dynamics. My potential for delivering breakthrough research is reflected by my results in MHD turbulence ('Uniturbulence', see first-authored publication no. 7 and 10), for which I was awarded the 2019 PhD Thesis Prize of The European Solar Physics Division (ESPD). I was the co-supervisor of a PhD student working on further characterising Uniturbulence who successfully defended their thesis in 2024. As a follow-up to the research done during my two-month stay (starting March 2022) with the Parker Solar Probe team at NASA/GSFC involving data analysis and comparison to 3D MHD numerical simulations of the solar corona and wind, I have successfully applied to lead a team at ISSI Bern, during which we (15 experts in the field) debated what is yet to be explained in the dynamics of the corona and solar wind in light of the Parker Solar Probe measurements, demonstrating my leadership capabilities.

Computational highlights

I have been using and developing various astrophysical MHD codes for almost the last ten years, including MPI-AMRVAC, FLASH, Pluto, Athena++ and KATHENA, simulating diverse plasma processes in the solar atmosphere and heliosphere. To date, I have used multiple Tier 2 and Tier 1 CPU and GPU systems at the KU Leuven and at the University of Warwick. I have successfully applied for computing times on two different Tier-1 systems of the Flemish Supercomputing Centre (VSC): On 'Breniac' in 2017: 1.000.000 CPU-hours, and on 'Hortense' in 2021: 2.360.000 CPU hours. I have implemented physics modules and capabilities in various finite-volume MHD codes, such as optically thin radiative losses in the FLASH code, modified spherical geometry and specialized heating functions in the MPI-AMRVAC code. My expertise in computational modelling is also demonstrated by requests to review proposals for runtime on some of the world's largest supercomputers. A summary of my previous published computational work (complete with animations from simulations) is available on: <https://magyarfnorbert.wixsite.com/home>

REFERENCES

- 1) Valery Nakariakov, University of Warwick, UK.
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2) Adam Szabo, NASA/GSFC, USA.

Email: adam.szabo-1@nasa.gov

Website: <https://science.gsfc.nasa.gov/sed/bio/adam.szabo-1>

3) Tom Van Doorselaere, KU Leuven, Belgium.

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